

Math 243: Homework 04

Due Friday, September 18

To receive full credit, it is necessary that you show your work.

Problem 1. Consider the curve C defined by the parametric equations

$$\begin{cases} x(t) = \cos(\frac{1}{t}) \\ y(t) = \sin(\frac{1}{t}) \end{cases}$$

for $1 \leq t < \infty$. Sketch this curve and compute the length of the curve.

Problem 2. Convert each of the following points from polar coordinates to Cartesian coordinates:

$$\begin{array}{ll} \left(-1, \frac{\pi}{6}\right) & \left(6, \frac{2\pi}{3}\right) \\ \left(5, \frac{7\pi}{8}\right) & \left(\frac{1}{2}, -\frac{5\pi}{6}\right) \end{array}$$

Problem 3. Convert each Cartesian coordinate to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

$$\begin{array}{ll} (2, 3) & (-1, 8) \\ (-1, -1) & (0, 6) \end{array}$$

Problem 4. In your own words, provide a geometric interpretation of the following identities:

$$x = r \cos(\theta), \quad y = r \sin(\theta), \quad \tan(\theta) = \frac{y}{x}, \quad x^2 + y^2 = r^2$$

Your explanations should include sketches.

Problem 5. Let C be the curve defined by the parametric equations

$$\begin{cases} x(t) = t^2 - t + 2 \\ y(t) = t - \frac{1}{3}t^3 \end{cases}$$

- (a) Find the equation of the tangent line to C at $t = 0$.
- (b) For what values of t is the curve convex up? For what values is it concave down?

Problem 6. Find $\frac{d^2y}{dx^2}$ at the point $(2, 0)$ for

$$x(t) = t^3 + 1 \qquad y(t) = t^2 - 1$$

Problem 7. Find the Cartesian equation of the graph $r = 4\cos(\theta)$ for $0 \leq \theta \leq \pi$, and sketch the graph.
Hint: you'll need to complete the square to convert to Cartesian form.

Problem 8. Find the tangent line to the curve at $t = 4$

$$x(t) = \sqrt{t} \qquad y(t) = t^2 - 2t$$

Problem 9. Consider the curve given by the parametric equations

$$\begin{cases} x(t) = t^{-3} \cos(t) \\ y(t) = t^{-3} \sin(t) \end{cases}$$

for $1 \leq t < +\infty$. Sketch this curve and compute the arc length.

Problem 10. Consider the polar function

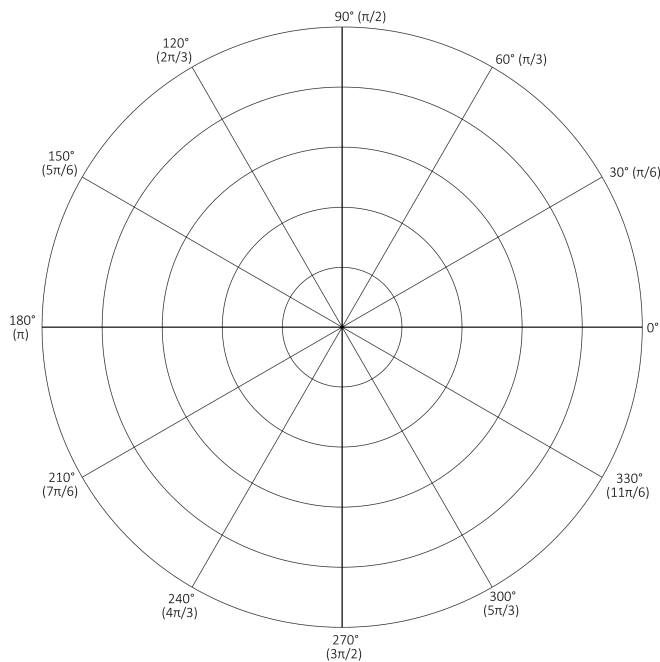
$$r = 1 + \sin(\theta)$$

where $-\infty < \theta < \infty$.

(a) Fill in the missing values in the following table.

θ	$r = 1 + \sin(\theta)$	θ	$r = 1 + \sin(\theta)$
$-\pi/2$		$\pi/2$	
$-\pi/3$		$2\pi/3$	
$-\pi/4$		$3\pi/4$	
$-\pi/6$		$5\pi/6$	
0	1	π	
$\pi/6$	$1 + \frac{1}{2} = 1.5$	$7\pi/6$	
$\pi/4$	$1 + \frac{\sqrt{2}}{2} \approx 1.7$	$5\pi/4$	
$\pi/3$		$4\pi/3$	
$\pi/2$		$3\pi/2$	

(b) Plot the points from part (a), and sketch the curve (also indicate the direction). You may wish to use a diagram like the following.



(c) Does the graph of the curve change if we restrict θ to only values $0 \leq \theta \leq 2\pi$? What about $0 \leq \theta \leq \pi$?

Problem 11. Consider

$$\begin{cases} x(t) = 1/t \\ y(t) = 1/t \end{cases}$$

for $1 \leq t < +\infty$. Sketch the parameterized curve. What is the arc length?